

SPEC - the green flow: cup, ball, putts, score, then the shots and their lies

Fable, 2026-09-16, at medium (Matt set it: /effort medium, "Gonahead"). Written for Opus to build at high (.claude/agents/opus.md). Fable reviews the diff against this file and signs in docs/DECISIONS_LOG.md.

His words that bound it, verbatim:

- 2026-09-13: "Is it possible for me to mark the cup, putt 1, and my score. Then the app computes the shot locations from the track and auto fill the lies and distances. Then I click on any entries to manually correct.?"
- 2026-09-15: "fix the lie card and yes the conformation when in question is needed when I am entering the score at the end of the hole. Workflow on the green mark the cup or my ball first whatever is easiest. Hole out - record the putt length for short putts, double check GPS for long putts, enter hole score (once this is entered the app needs to compute the shots and ask me questions about the lie. Shot 2 rough or fairway, shot 3 green or fairway, etc..."
- 2026-09-16: "With the Green flow established at 20ft not 15" (ruling in docs/DECISIONS_LOG.md, commit a778194).
- Standing, 2026-09-13: on the course he touches cup, putt 1 and score, no tee marks (docs/HANDOFF-native-build.md Section 3). "If there is a penalty I will log it when it occurs and the score is adjusted after holing out on that hole."

Provenance tags: [measured] read from code this session, [design] this spec's decision, [verify] Opus confirms during the build.

0. What this is, and what it is not

Is: one sheet on the green that runs his whole hole in his order - mark the cup and the ball in either order, hole out, the putts (typed under 20 ft, GPS at 20 ft and over), the score, and then the app proposes the full shots out of the track and asks him the lie of each. Every piece of it exists today in js/ui/screen-play.js [measured]; this job joins them into one path and applies the 20 ft rule.

Is not: a change to the mark burst or the lie card (v25-v27, field-fit), to the shot ranking in js/round/track-analysis.js, to the round record (schemaVersion unmoved, no new keys), to the export, or to the hint strings whose fit REPORT_2.6 measured. Lies from the course map are NOT in this job: the course map as app data is its own xhigh spec on his word, and until it lands every lie is his tap (NOT SURE flags the app's guess, as today). The "in question" edge band from docs/HANDOFF-native-build.md Section 4 item 6 waits for that map.

1. What exists today [measured]

Piece	Where	State
MARK CUP burst, saveCup, reopens the putt sheet	screen-play.js 1096-1147	done
The putt sheet: putt count, feet grid per putt, GPS for putt 1 (ball mark to cup) with its error bar, pin sheet when no cup, SAVE	openGreenEntry, 1206-1640	done
A ball mark on the green	only through MARK SHOT n + lie GREEN (the lie card)	the pocket flow has no way to mark the ball
The score card (strokes, putts, penalties, first putt grid)	openHoleCard, 2716-2810	done, reached only by END-OF-HOLE ENTRY
The shots proposed from the track and the lie asked per shot, NOT SURE, NOT A SHOT, PLAYED TWICE, the cup from the track	openShotConfirm, 2823-2990	done
Writing the hole	applyHoleEntry, 3008-3060	done; drops a marked ball (h1.shots = [], item 4 below)
The 20 ft rule	nowhere	this job
The capture kind putt	commit has no branch for it; the burst label "Marking putt" exists (717)	this job

holeWindow (js/round/round.js 660-750) already tolerates this hole's own cup and ball marks: own marks only lower fromTs and raise toTs, never narrow the window [measured]. setGreenEntry already preserves a marked first putt (markedFirst) [measured].

2. Decisions [design] (each overridable by Matt, in his chat)

#	Decision	Why
G1	The 20 ft rule. With a GPS first-putt distance (ball mark to cup) of 20 ft or more , GPS is the value and the sheet shows it with its error bar for him to check; a typed value is optional. Under 20 ft, or with no GPS distance , the typed distance is required before the sheet saves. The constant is TYPED_PUTT_MAX_FT = 20 in one place with his words beside it.	His ruling. Cost table in docs/HANDOFF-native-build.md Section 7: an 8 ft error costs 0.17 strokes at 20 ft, 0.34 at 15 ft.
G2	MARK BALL is a burst of kind putt, from inside the sheet, no lie card. It saves a green shot with the mark (or sets the mark on the existing first green shot) and reopens the sheet, the way MARK CUP does. RE-MARK BALL after.	"mark the cup or my ball first whatever is easiest": both are one tap from the same sheet, either order. The lie is known - it is the green.
G3	After the putts, the score, then the shots - on a hole with no full-shot marks. SAVE on the sheet goes straight into the score card with putts and the first putt carried in (not asked twice), then FIND MY SHOTS, then the confirm stage, then SAVE HOLE. On a hole that has full-shot marks, SAVE behaves as today: the marks are the shots.	His order: putts, then score, then the questions. The pocket flow is the intended way to play (rev 2); the marked flow stays for the days he marks.
G4	The hole is written once, at SAVE HOLE, and the ball and cup marks survive it. applyHoleEntry keeps green-lie shots that carry a mark, so the first putt keeps its measured position.	The ball mark is what locates the first putt for GPS (G1) and the hole for every earlier shot; dropping it would silently turn a measured putt into an unmeasured one.
G5	One tap out at every step. Leaving the score card or the confirm stage keeps the putts already saved; the hole stays in_progress and everything on it is editable later from the hole menu as today. Nothing here blocks NEXT HOLE.	Design rule: no app-driven state may block logging reality; the golfer is the source of truth.
G6	No new schema. A GPS-chosen first putt is stored as today (no distanceFt; firstPuttM measures mark to cup). A typed one goes through setShotDistance in feet. greenEntry unchanged.	The round rails are proven; provenance already distinguishes typed, GPS and track-inferred.
G7	The footer does not grow. ENTER PUTTS ☐ is relabelled GREEN ☐ (one word fits at 360 px [verify]); END-OF-HOLE ENTRY ☐ stays for a hole entered from the next tee or later. The fold tests of v25-v27 stay green untouched.	REPORT_2.6: every footer pixel was fought for.

3. The flow, step by step (his hole, phone in the pocket until the green)

- Tee to green:** nothing. The track records.
- On the green:** GREEN ☐ opens the sheet (or it opens itself after a MARK CUP from the footer, as today). The sheet's head row holds two buttons: **MARK CUP** / RE-MARK CUP and **MARK BALL** / RE-MARK BALL. Either order. Each is the existing 3 s burst; the sheet reopens after it. The pocket lock and the hints are unchanged.
- Hole out.** In the sheet: putts (segmented 0-4+, default 2, as today). Under **Putt 1 - to the hole:**
 - GPS distance available (ball mark and cup): show it as today ($N \text{ ft} \cdot \text{GPS} \pm E$).
 - $\geq 20 \text{ ft}$: GPS is selected; the grid is there to override; SAVE is enabled. Label under the readout: *"20 ft and over: GPS stands. Tap a number only if you know better."*
 - $< 20 \text{ ft}$: the grid is required; the readout shows the GPS number greyed with *"GPS says N ft - under 20 ft, type it"*; SAVE stays disabled until a value is typed for putt 1 (0 putts needs nothing).
 - No GPS distance: the grid is required, as the $< 20 \text{ ft}$ case; the pin sheet control stays where it is for a hole with no cup.
 - Putts 2+ unchanged.
- SAVE.** setGreenEntry as today (the marked ball preserved). Then:
 - strokeMarks(h1).length > 0 (he marked shots): done, as today.

- otherwise the **score card** opens with putts and firstPuttFt carried from the sheet and not shown again: **Strokes** (default $\max(\text{par}, \text{putts} + 1)$), **Penalty strokes** (default 0). One line under it: *"Score minus putts minus penalties is what the track looks for."* FIND MY SHOTS □.
5. **The shots.** proposeHoleShots on holeWindow, the confirm stage as today: each shot's card, lie segmented (Tee/Fairway/Rough/Sand/Recovery), NOT SURE, NOT A SHOT / PLAYED TWICE; the cup from the track only if no cup was marked. **SAVE HOLE** → applyHoleEntry (G4) → toast with UNDO as today → the footer's next action is NEXT HOLE.
 - No track in the window: the existing "No track for this hole" sheet, HAND-ENTER INSTEAD. The putts stay saved.
 6. **Corrections later:** the hole menu and EDIT PUTTS as today; re-running the flow on a saved hole goes through the same sheets and replaces the hole (UNDO stands).

4. Web-side changes, file by file (surgical; nothing else moves)

1. js/ui/screen-play.js. (a) const TYPED_PUTT_MAX_FT = 20; near CUP_AT_TEE_M (1094) with his words. (b) beginCapture('putt'): commit gains a kind === 'putt' branch → saveBall(h1, reduced): the first green shot with no mark gets mark from reduced (same shape addShot stores), else addShot(h1, { lie: 'green', reduced, source: 'gps', club: 'putter' }); noteMark('Ball marked here. '); the poor-accuracy warning as for the cup; persist; reopen the sheet (reopenPuttsAfterCup pattern, renamed reopenPuttsAfterMark and used by both). No learnGreen call from here [design]: that accumulator learns from the lie card's marks today and a second feeder is a separate decision. (c) openGreenEntry: the MARK BALL / RE-MARK BALL button beside the cup control (both in one btn-row at the top of the sheet, above the putt count); the G1 rule in the putt 1 field and on SAVE's disabled; SAVE's handler then calls openScoreCard(h1) when strokeMarks(h1).length === 0. (d) openHoleCard becomes openHoleCard(h1, { fromSheet }): with fromSheet the putts and first putt fields are hidden and prefilled from h1 (holePutts, puttDistancesFt(h1)[0]); the existing END-OF-HOLE path passes nothing and is unchanged. (e) applyHoleEntry: keep h1.shots.filter((s) => s.lie === 'green' && s.mark) through the rebuild (G4); setGreenEntry then finds markedFirst as it does today. (f) The footer button text ENTER PUTTS ▶ → GREEN ▶; nextStepHint strings untouched.
2. js/round/round.js. No change unless (e) needs a helper; if so a pure greenMarks(hole) beside holePutts.
3. js/data/schema.js, js/data/store.js, js/round/track-analysis.js, css/*, index.html: no change.

5. What must not change

js/gps/gps.js, js/round/track-analysis.js, the lie card and its fold (paint, the card-first body order, the $\min\max(0, 1fr)$ grids), the two hint strings of v27, the mark burst's timing, schemaVersion, formatVersion, REVISION, BUILD.id (the bump is the deploy's, on his word), the pocket lock, js/native/*, android/.*.

6. Tests (/test/, each proven to fail without its change)

1. **The 20 ft rule, pure.** A helper firstPuttEntryMode({ gpsFt }) (or the sheet's own logic exposed for the test) returns gps at 20 and 60, typed at 19.9, 8 and null.
2. **The sheet at 360x728 with a ball mark 30 ft from the cup:** the putt 1 readout shows the GPS number, SAVE is enabled with nothing typed, and the saved hole's first putt has no distanceFt and firstPuttM equals the mark-to-cup distance.
3. **The same sheet at 12 ft:** SAVE is disabled until a value is typed; after typing 10, puttDistancesFt(h1)[0] === 10.
4. **MARK BALL from the sheet:** the burst of kind putt saves one green shot with a mark and no lie card is shown; a second MARK BALL replaces the mark rather than adding a shot.
5. **The score card after SAVE on an unmarked hole** opens with the putts and first putt carried in and not shown; with a marked tee shot SAVE closes the sheet and no card opens.

6. **SAVE HOLE keeps the ball and the cup:** after `applyHoleEntry` on a hole with a ball mark and a cup, the first green shot still carries its mark and `h1.cup` is unchanged.
7. **The v25-v27 fold group stays green** as it is (no new assertion; the run proves the footer did not grow).

Existing suite: 531 at v27 + shell; all stay green.

7. What Matt decided, and what is his to decide

- Decided: 20 ft (a778194). Cup, putt 1, score is the on-course contract.
 - His, before the build if he wants, else Fable's defaults stand: the footer label **GREEN** ☐ (G7); the score card's default strokes `max(par, putts + 1)` (G3).
 - Not this job, his word later: lies from the course map (the xhigh spec), the auto-lie edge band, `learnGreen` from the ball mark.
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8. Order of work for Opus (one commit per step is fine)

1. `TYPED_PUTT_MAX_FT`, the sheet's putt 1 rule and SAVE gating; tests 1-3.
2. `MARK BALL` (putt capture, `saveBall`, `reopen`); test 4.
3. `SAVE` → score card → confirm → `SAVE HOLE` with the marks kept; tests 5-6; footer label; the fold group re-run (test 7).
4. `BUILD.id` is NOT bumped by Opus (Fable bumps at the push, on his word). `docs/handoff/REPORT_3.2.md` + PDF (`tools/md2pdf.py`): every number with n, the diff list, any departure from this spec and why, and what he does differently on the green in five lines.

Return `DONE <hash>` or `BLOCKED <question>`. A spec line that cannot work as written is a `BLOCKED` with what was found, not a quiet variation.